ISM 4117-100
Data Mining & Predictive Analytics

Fully Online
3 credits – CRN 13400

SPRING, 2026
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Course Description

This course introduces the core concepts of data mining (DM), its techniques, implementation and benefits. It also identifies industry branches that benefit most from Data Mining, such as retail, target marketing, fraud protection, health care and science and web and e-commerce. Detailed case studies and using leading mining tools on real data are presented.

Instructional Method

This class is designated as an asynchronous fully online course. Students are expected to read assigned chapters, engage actively in online discussions, complete assignments by specified deadlines, and reach out to the instructor for clarification or assistance as needed. Successful participation and timely completion of coursework are key components for a rewarding learning experience in this fully online asynchronous course. Lectures are recorded and posted according to the schedule.

Please note: In an asynchronous course, there are no fixed class times. Instead, students have the flexibility to access course materials, lectures, and assignments at their own pace within specified deadlines. This format allows for greater flexibility, enabling students to manage their studies around personal schedules and commitments.

Prerequisites/Corequisites

No course prerequisites

Course Objectives/Student Learning Outcomes

Provide the students with an understanding of what they will be expected to learn as a result of successfully completing the course.

The primary aim of this course is to acquaint students with the practical challenges that are encountered when solving real-world data mining problems. By allowing students to encounter the complete picture of going from business level problem formulation, through the analysis of quantity and quality of available data, as well as selection of suitable algorithms and evaluation of the obtained results. Students are expected to learn not only about the advantages of various data mining and predictive analytics methods, but also about their limitations. The focus in this course will be on applying, in practice, knowledge, and concepts from business models and on understanding the applicability of data mining and predictive analytics.

Students will reinforce the learning of data mining concepts by means of data analysis techniques to make better business decisions through proper data preparation, data exploration, and simple tools for solving data mining problems. Students will be introduced to advanced concepts such as data mining applications, predictive analytics, web mining, text mining, and ethical aspects of data mining and predictive analytics. Students will learn to mine heterogeneous data and demonstrate proficiency in classification and prediction applications such as neural networks, linear regression, cluster analysis, market basket analysis, and decision trees.

Working as a team (2-4), students will demonstrate proficiency in applying data mining analytical techniques on an advanced real-world business problem that examines a large amount of data to discover new information in addition to analyzing and evaluating technique effectiveness with a less than perfect constantly evolving technology by presenting a self-designed semester project. Commencing with several singular technique projects and concluding with the comprehensive semester project, students will reinforce their oral skills by way of presentations as well as written and critical thinking skills by the use of executive memos requiring quantitative analysis and evaluation.

Course Evaluation Method

In-class problems (10)	20%	20%
Exam 1	20%	60%
Exam 2	20%	
Exam 3	20%	
Team Final Project and Presentation	20%	20%
Total	**100%**	**100%**

Exams: The exams (midterm exams and final) will include True/False questions and data mining application problems, administered on Canvas and will cover the content from the material presented in the lectures, assignments and the data mining techniques using XLMiner.

- Usually, students will be asked to interpret results from applying a specific data mining method, such as confusion matrices and classification false positive/negative rates.
- **Recorded Lectures, Class Problems, Discussions and good notetaking are essential elements for success.**
- **Class Problem:** This course features ten hands-on problems.
    - I'll guide you through solving each one and provide recorded solutions, along with step-by-step instructions.
    - The exercises are designed to help you understand class concepts and gain practical experience in using XLminer for data modeling.

Course Grading Scale

Grading Scale:

A	93.00-100%	**C**	73-76.99%
A-	90-92.99%	**C-**	70-72.99%
B+	87-89.99%	**D+**	67-69.99%
B	83-86.99%	**D**	63-66.99%
B-	80-82.99%	**D-**	60-62.99%
C+	77-79.99%	**F**	< 60 %

- Note that this is a "core" business course for which a grade of 73 "C" or better is required to pass the course.

Final Project:

- The final project will demonstrate a comprehensive understanding of the course. It requires locating a large data set (typically with over 1000 records) with at least 8 variables of different data types, preparing and understanding the data, and addressing a business question suitable to the data chosen.

- Project teams will consist of 3-4 students. Team partners will be assigned. Typically, in real work settings you do not get to choose who you work with, so it is important that you learn to work well with others, even if you don't always have a lot in common. If there are differences between who you work with, it is your responsibility to try and resolve them. If you are unable to then please let me know and other arrangements will be made. It is to your best advantage to document (email) your communications to avoid complications; if you feel more comfortable, use Google Docs to complete your project and/or feel free to cc: your emails to me.

- While you will collaborate with your peers on the team projects, each team member must contribute an equal portion of effort or work towards each phase of the project. Based on observations of group work by the professor, team members contributing more than their fair share may receive higher grades than other group members. It is the group's responsibility to advise the professor of any issues or challenges that may hinder your ability to complete any phase of the project.

- **The project submission must include the following:**
    - The actual Excel spreadsheet file(s) where the method was applied.
    - The necessary additions such as confusion matrices, classification rates, etc., that reinforce the appropriate conclusions (can be added as worksheets to the original Excel file).
    - A PowerPoint deck that concisely presents, summarizes, and analyzes the results (draw meaningful conclusions and comparisons).
    - A recording of your team presenting your project. You can send the .mp4 file or recording (recommend using Zoom) link.

Policy on Makeup Tests, Late Work

- It is important that each exam/problem be submitted by the scheduled time and date.
- **Late class problems can be submitted after the due date until May 5th , 2025. Each late assignment will be assessed a 25% penalty.**
- Assignment schedules are provided are available with ample time for submission, the student is responsible for planning their schedule and work. Submitting an assignment late because of technical challenges will not excuse you from a late submission penalty, ample time is provided to submit assignments.
- An exam/problem missed without prior approval and verifiable documentation that the unapproved absence was unavoidable as stated above does not qualify to be made up.

Late Project:

- Grade penalty equal to 15% of the project grade per day late will be applied after the project's due date. Late submission 5 days after the due date will receive 0 points.

Participation Policy

- Welcome to this online asynchronous course! I believe that learning is a collaborative process, and your success in this course is dependent on your active participation in our virtual learning community. Therefore, it is important to fully engage with the course materials and complete all assignments and activities.

- Throughout this course, any changes to the schedule will be communicated to you through our course platform, so it is essential to regularly check for updates. Occasionally, there may also be practice problems or exams. Please understand that exams cannot be made up if missed, so it is crucial to stay on track and complete them as scheduled.

- As you progress through the course, it is important to keep up with the materials. The content is cumulative, meaning that each module builds upon the previous ones. To ensure that you grasp the concepts effectively, it is highly recommended that you dedicate regular time to study the assigned materials and actively participate in the provided exercises and discussions.

- Please note that failing to engage with the course materials and assignments will have a negative impact on your performance. Regularly studying the assigned materials, and completing exercises in a timely manner are key factors for successfully passing the assessments in this course. Therefore, we strongly encourage you to prioritize your learning, allocate sufficient time, and actively participate to maximize your learning experience and increase your chances of successfully completing this course.

Classroom Etiquette/Netiquette Policy

- Each student is responsible for keeping up with the class schedule, checking your FAU (Florida Atlantic University) email account, and checking the course Canvas site on a regular basis.
- If you use a non-FAU email address as your primary address, arrange for FAU email to be forwarded.
- Please use email to contact me.

Anti-plagiarism Software

- Written components of any assignment or project may be submitted to anti-plagiarism software to evaluate the originality of the work. Any students found to be submitting work that is not their own will be deemed in violation of the University's honor code discussed above.

Student Responsibilities

- Students are responsible for reading the required material (designated chapters) as scheduled.
- Students are responsible for reviewing the class schedule provided on Canvas often to know important course due dates (Tests, quizzes, etc.).
- Students are responsible for reviewing all material covered and assignments/exams as scheduled or by announcement if there are any changes during their absence.

Selected University and College Policies

Expectations for Students

- **Announcements -** You are responsible for reading all announcements posted by the instructor. Check the course announcements each time you log in.
- **Email/Video Conferencing -** You are responsible for reading all your course email and responding in a timely manner.
- **Course-Related Questions -** Post course-related questions to the FAQ Discussion board. This allows other participants with the same question to benefit from the responses. Also, make sure you review this forum prior to posting a question. Someone may have already asked and answered the question in previous posts.

Instructor's Plan for Classroom Response Time & Feedback

- **Email/Video Conferencing Policy -** Except for weekends and holidays, the instructor will typically respond to email (Canvas inbox or FAU email) in a timely manner. If you have questions of a personal nature, you should email the instructor. As a student, you are responsible for reading all your course email and responding in a timely manner.
- **Assignment Feedback Policy -** The instructor will provide feedback on submitted assignments within one week of the submission date. Some assignments may require a longer review period, which the instructor will communicate to you.
- **Assignment Feedback Method -** The instructor will provide feedback on submitted assignments in a timely manner. Some assignments may require a longer review period, which the instructor will communicate to you.
- **Course-Related Questions Policy -** Except weekends and holidays, the instructor will generally answer questions within a timely manner.
- **Electronic Communication Policy**
- **In addition to the University's policy, please consider the following:**
- Privacy, confidentiality, and security in all electronic communications.
    - All electronic communication resources must be used for the course and in alignment with the University mission.
    - Prohibited use of false identity, false identity pseudonyms, or anonymous (sender's name or electronic identification is hidden).
    - Access without consent.
- Disruption of services including introducing computer contaminants (viruses).
- Harassment of any kind.

Please see the Office of Information Technology's policies on Cyber Security Awareness.

Counseling and Psychological Services (CAPS) Center

Life as a university student can be challenging physically, mentally and emotionally. Students who find stress negatively affecting their ability to achieve academic or personal goals may wish to consider utilizing FAU's Counseling and Psychological Services (CAPS) Center. CAPS provides FAU students with a range of services – individual counseling, support meetings, and psychiatric services, to name a few – offered to help improve and maintain emotional well-being. For more information, go to http://www.fau.edu/counseling/

Disability Policy

In compliance with the Americans with Disabilities Act Amendments Act (ADAAA), students who require reasonable accommodations due to a disability to properly execute coursework must register with Student Accessibility Services (SAS) and follow all SAS procedures. SAS has offices across three of FAU's campuses – Boca Raton, Davie and Jupiter – however disability services are available for students on all campuses. For more information, please visit the SAS website at www.fau.edu/sas/.

Code of Academic Integrity

Students at Florida Atlantic University are expected to maintain the highest ethical standards. Academic dishonesty is considered a serious breach of these ethical standards, because it interferes with the university's mission to provide a high-quality education in which no student enjoys an unfair advantage over any other. Academic dishonesty is also destructive of the university community, which is grounded in a system of mutual trust and places high value on personal integrity and individual responsibility. Harsh penalties are associated with academic dishonesty. For more information, see University Regulation 4.001.

Withdrawals

Any student who decides to drop out is responsible for completing the proper process required to withdraw from the course.

Grade Appeal Process

A student may request a review of the final course grade when s/he believes that one of the following conditions apply:

- There was a computational or recording error in the grading.
- Non-academic criteria were applied in the grading process.
- There was a gross violation of the instructor's own grading system. The procedures for a grade appeal may be found in FAU Regulation 4.002.

Disruptive Behavior Policy Statement

Disruptive behavior is defined in the FAU Student Code of Conduct as "*... activities which interfere with the educational mission within classroom*." Students who behave in the classroom such that the educational experiences of other students and/or the instructor's course objectives are disrupted are subject to disciplinary action. Such behavior impedes students' ability to learn or an instructor's ability to teach. Disruptive behavior may include but is not limited to non-approved use of electronic devices (including cellular telephones); cursing or shouting at others in such a way as to be disruptive; or other violations of an instructor's expectations for classroom conduct.

Policy on the Recording of Lectures

Students enrolled in this course may record video or audio of class lectures for their own personal educational use. A class lecture is defined as a formal or methodical oral presentation as part of a university course intended to present information or teach students about a particular subject. Recording class activities other than class lectures, including but not limited to student presentations (whether individually or as part of a group), class discussion (except when incidental to and incorporated within a class lecture), labs, clinical presentations such as patient history, academic exercises involving student participation, test or examination administrations, field trips, and private conversations between students in the class or between a student and the lecturer, is

prohibited. Recordings may not be used as a substitute for course participation and may not be published or shared without the written consent of the faculty.

Faculty Rights and Responsibilities

Florida Atlantic University respects the right of instructors to teach and students to learn. Maintenance of these rights requires classroom conditions which do not impede their exercise. To ensure these rights, faculty members have the prerogative:

- To establish and implement academic standards
- To establish and enforce reasonable behavior standards in each class
- To refer disciplinary action to those students whose behavior may be judged to be disruptive under the Student Code of Conduct.

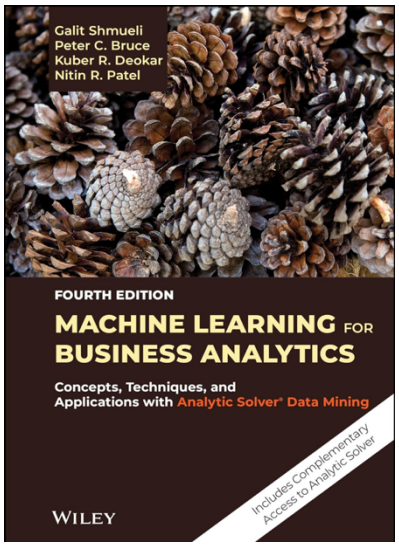
Required Materials:

- A copy of Microsoft Office
    - Microsoft Excel will be used to do calculations, data analysis, and graphing
    - Microsoft Office 365 is available for free for all currently enrolled FAU students
    - For more information, visit http://www.fau.edu/oit/getoffice365

- **Data Mining Software: (Required)** XLMiner® (Analytic Solver Platform) for Windows - a comprehensive data mining add-in for Excel, with neural nets, classification and regression trees, logistic regression, linear regression, Bayes classifier, K-means, association rules, clustering, and principal components analysis. Students can download the XLMiner on their personal computers **using instructions posted on Canvas.**
- **The course code is: CRN13400JF26SP**
- **The book code is: LABBYOD**
- **Support@solver.com - Support for XLMiner.**
- A Windows-based PC is preferred; Macs are fine for the most part, but the commands can be slightly different. I use a Windows-based PC. Please let me know as soon as possible if you do not have access to a Windows-based PC.
- Additional materials will be assigned and distributed via Canvas throughout the course.

Optional Texts/Readings

- **Title:** Machine Learning for Business Analytics: Concepts, Techniques, and Applications with Analytic Solver Data Mining (4th Edition)
- **Author:** Galit Shmueli, Peter C. Bruce, Kuber R. Deokar and Nitin R. Patel
- **ISBN:** 97811198298367
- **Publisher:** Wiley



Competencies

- Know how to use & navigate the Canvas Learning Management System
    - Canvas Student Getting Started Guide
    - Canvas Video Guides for Students
- Set up your notification options on Canvas.
- Students should have a working knowledge of basic math (algebra) and Microsoft Excel.
- Students should have access to Excel spreadsheet software, and students are assumed to be familiar at an intuitive level with general business practices of collecting, storing, and using data.

Course Schedule: **Updated: 12/23/25 - Class Week is Monday to Sunday**

Start	End	Topic	Self-Reading
1/12/2026	1/18/2026	Introduction - Basic Data Mining Techniques	Ch.1 - Ch.2
		1/16/2025 - Last day to drop a course without consequences	
1/19/2026	1/25/2026	Data Visualization	Ch.3
1/26/2026	2/1/2026	Dimension Reduction	Ch.4
2/2/2026	2/8/2026	Evaluation of Model Performance	Ch.5
2/9/2026	2/15/2026	*Exam 1 Ch. 3,4,5 - Due Feb 15th*	
2/16/2026	2/22/2026	Linear Regression	Ch.6
2/23/2026	3/1/2026	Logistic Regression	Ch.10
3/2/2026	3/8/2026	k-Nearest-Neighbors (Assignment Due Date Extended)	Ch.7
3/9/2026	3/15/2026	Spring Break March 7th to March 13th	
3/16/2026	3/22/2026	*Exam 2 CH. 6,10,7 - Due Mar 22nd* **Mar 27th - Last Day to drop with a "W"**	
3/23/2026	3/29/2026	Neural Networks Data Set and Business Questions Due Mar 29th	Ch.11
3/30/2026	4/5/2026	Classification and Regression Trees	Ch. 9
4/6/2026	4/12/2026	Cluster Analysis	Ch.16
4/13/2026	4/19/2026	Association Rules – Final Project Draft Due Apr 12th	Ch.15
4/20/2026	4/26/2026	*Exam 3 – Ch. 11, 9, 16, 15 - Due Apr 26th* *Team Final Projects Due Apr 20th*	
4/27/2026	5/2/2026	**All Late Assignments Due – May 5th**	
		Final Grades Submitted – May 11th 9AM	

The instructor reserves the right to adjust this syllabus as necessary.